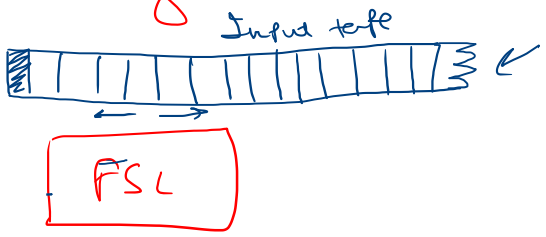
Turing Machine



$a^n b^n c^n : n \geq 0$

$\{a^n b^n c^n d^n e^n f^n g^n h^n \ldots\ldots\ldots\ldots\ldots\}$

Lewis Papadimitriou

Chapter 4 (TM)

Salient features of TM :-

1.) It has a bidirectional memory as the input tape
2) The machine can stay at its place after reading from a cell
3). It's a read write tape.
4). The tape is left ended, but may be extended to ∞ on the right
5). Initially each cell of tape has a special Element called "blank"

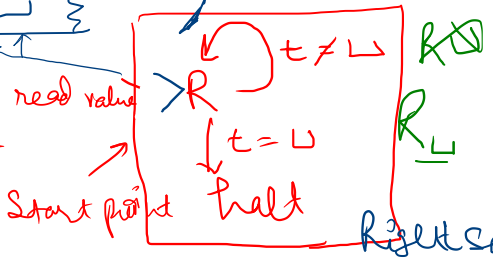


R → move one cell right

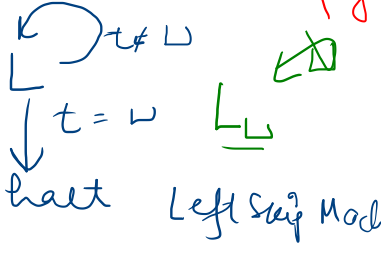
L → " " " left

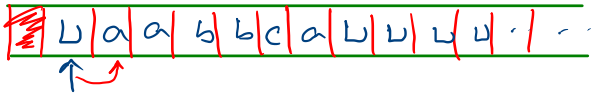


Current config = ⊔ W ⊔ ⇒ ⊔W⊔

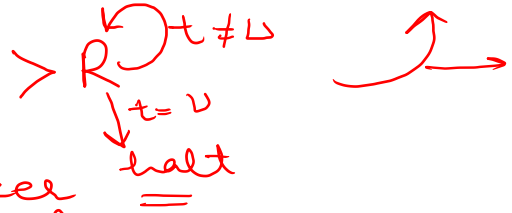


new config





Q1 a TM that skips all Non blank elements & halts at the FIRST blank on the Right.



marker for initial head position

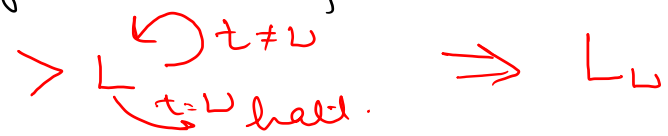
① $R_{⊔}$

② $L_{⊔}$

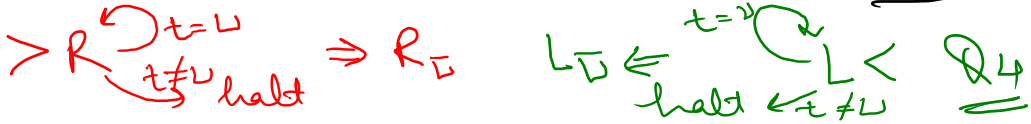
③ $R_{\bar{⊔}}$

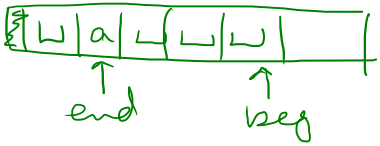
④ $L_{\bar{⊔}}$

Q2 Design a TM for ⊔W⊔ ⇒ ⊔W⊔ ?

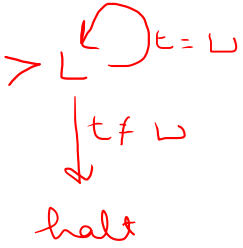


Q3. Design a TM which halts at FIRST Non blank on the RIGHT?

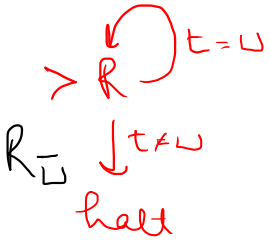




$\rightarrow R_{\sqcup}, L_{\sqcup}, R_{\bar{\sqcup}}, L_{\bar{\sqcup}}$ ← R → Right move



$L_{\bar{\sqcup}}$

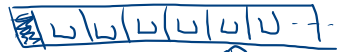


$R_{\bar{\sqcup}}$

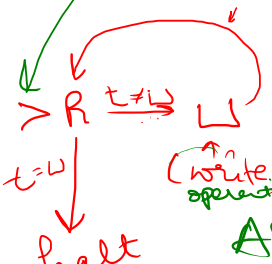
Q5



lt Deletion M/c



← this arrow doesn't change the head pos.



rt Deletion M/c

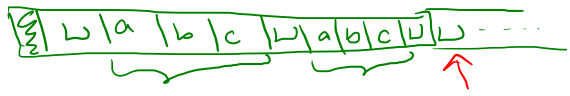
replace by ⊔

A TM that deletes towards right

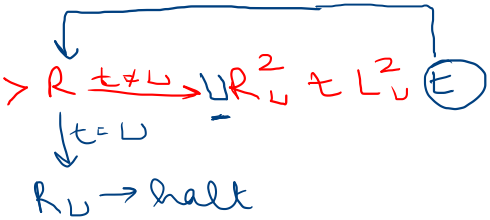
Q6



Copy machine



$\sqcup W \sqcup \Rightarrow \sqcup W \sqcup W \sqcup$



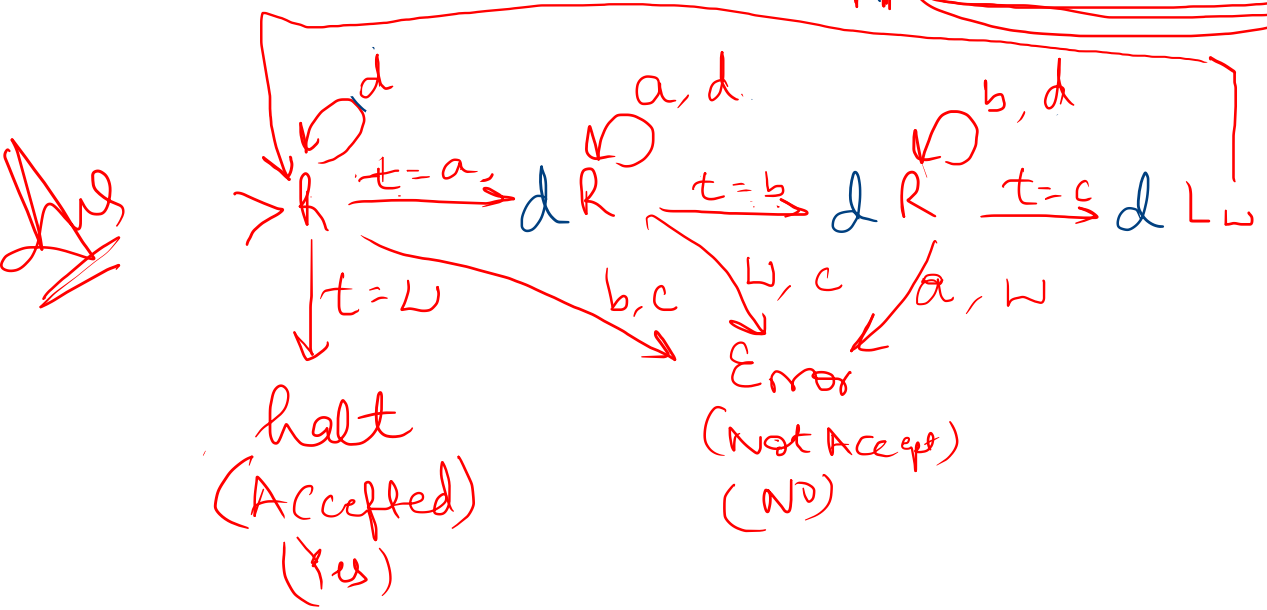
Copy machine

TM can be designed

1. Special String operations cut, copy, paste, clear, ...
2. Decision problems $a^n b^n$, $a^n b^n c^n$ ...
3. Math functions a+b, a−b, a×b ...

TM for $a^n b^n c^n$ $n \geq 0$



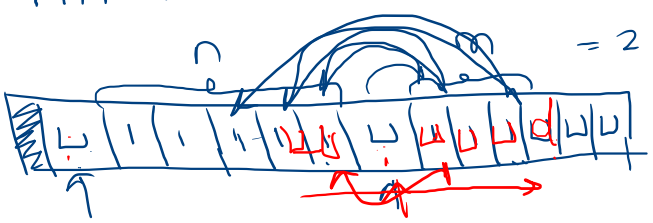


Turing m/c for Math functions

1. $f_1(n,m) = n+m$
2. $f_2(n,m) = n-m$

B, D, 0, H

Unary 1111111111



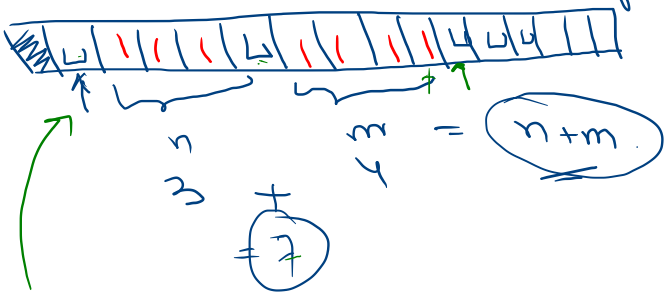
$> R_\sqcup R_{\bar{\sqcup}} \sqcup L_{\bar{\sqcup}} \sqcup$

$\Rightarrow > R_\sqcup^2 d L_\sqcup R_{\bar{\sqcup}} \sqcup L_{\bar{\sqcup}} \sqcup$

$t = d$

halt

eg. numbers n, m kept

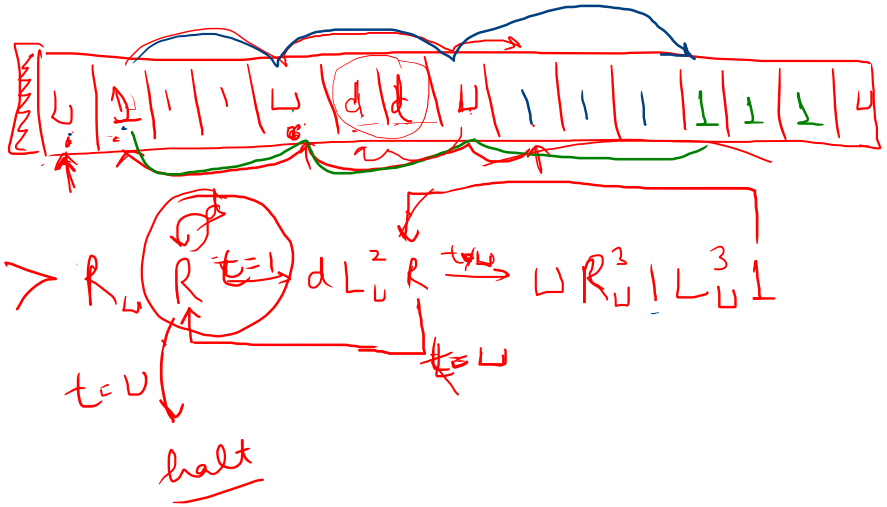


$> R_\sqcup 1 R_\sqcup L_\sqcup$ Ans

3] $f_3(n,m) = n*m$



copy 'n' 'm' times after m on the tape.



Primitive Recursive fn

Composition
1. Zero fn :- $Z(n) = 0$
2. Successor fn :- $S(n) = n+1$
3. Projection fn :- $U_2^3(x, y, z) = y$

Recursive :- $f(x_1, x_2, x_3 \dots x_n, y)$ $g(n)$, $h(n+2)$ such that

$$f(x_1, x_2 x_3 \dots x_n, 0) = g(x_1, x_2 x_3 \dots x_n)$$

$$f(x_1, x_2 x_3 \dots x_n, y+1) = h(x_1, x_2 x_3 \dots x_n, y, f(x \dots y))$$

$f_1(n, m) = n + m$ $\quad g(n)$, $h(n, m, f(n, m))$

$f_1(n, 0) = n + 0 = n \Leftarrow U_1^1(n) \Leftarrow g(n)$

$f_1(n, m+1) = n + m + 1 = S(n + m) \Rightarrow S(f_1(n, m)) \leftarrow h(\,)$

$\Rightarrow S(U_3^3(n, m, f_1(n, m)))$

hence Proved

$f_2(n, m) = n * m$ $\quad g(n)$, $h(n, m, f_2(n, m))$

$\downarrow$

$f_2(n, 0) = n * 0 = 0 \Leftarrow Z(n)$

$f_2(n, m+1) = n(m+1) = nm + n \Rightarrow f_1(n * m, n)$

$\Rightarrow f_1(f_2(n, m), n)$

$\Rightarrow f_1(U_3^3(n, m, f_2(n, m)), n)$

$\Rightarrow f_1(U_3^3(n, m, f_2(n, m)), U_1^3(n, m, f_2(n, m)))$

$f_3(n, m) = n^m$

$f_3(n, 0) = n^0 = 1 \Leftarrow S(Z(n))$

$f_3(n, m+1) = n^{m+1} = n^m * n \Rightarrow f_2(n^m, n)$

$\Rightarrow f_2(f_3(n, m, n))$

Hence proved

$\Rightarrow f_2(W_3^3(n, m, f_3(n, m)), W_3^3(n, m, f_3(n, m)))$

$f_4(n, m) = 3n^2 + m * n^2$

$P_2(P_1, I_1)$ given I_1 to P_1

P_1 will always halt

P NP NPC NPH

P

Det TM

[Polynomial time